Arthur Bauville

Yokohama, JP | a.bauville@gmail.com
Permanent resident visa

Senior software engineer / data scientist

linkedIn | github | portfolio | scholar

Skills

Programming languages: Python, C, SQL, MATLAB, Shell, GLSL, JavaScript/HTML/CSS
Libraries and tools: Numpy, Scipy, PyTorch, OpenCV, OpenGL, Amazon AWS
Techniques: finite element, finite difference, smooth particle hydrodynamics, arbitrary eulerian-lagrangian schemes, computational fluid dynamics, machine learning, optimization, deep learning
Project management: SCRUM, git, GitHub projects.

Experience

Axelspace – Tokyo, Japan

Senior ***remote sensing engineer*** *July 2024 – Present*

- (2026 -) Main developer and maintainer of the satellite image processing software
- (2025 - 2026) Led a team of 4-7 engineers. Implemented a modified SCRUM workflow that maximized face-to-face discussion during our weekly office day and managed tasks using GitHub Projects. I enjoyed some aspects of management, but I prefer engineering and research work.
- Supervised the release of a new image processing pipeline. Fixed initial stability through a fast iteration loop between Quality control and Development teams.
- Designed and implemented a new geometric image registration algorithm that combined prior knowledge and error filtering that reduces the 99% error from 125 m to 18 m.

Earth observation data scientist *July 2022 – June 2024*

- Designed a fast color-matching scheme that ensures continuity across cell boundaries. It allowed for satellite image mosaic of Japan to be computed in parallel (>1 TB data).
- Trained and fine-tuned deep learning models (PyTorch, HuggingFace) for semantic image segmentation.
- Supervised and mentored junior members and dispatch workers.

Japan agency for marine-earth sciences and technology – Yokohama, Japan

Researcher in the Mathematics and Advanced Technology group *Jan 2016 – Mar 2022*

- Developed alone a 2D multiphysics simulation code from scratch (C, Python, OpenGL, OpenMP; >50,000 lines of code). The software employs a finite-difference solver on a staggered grid to simulate Stokes flow with non-linear elasto-visco-plastic rheology and porous flow (i.e., two-phase flow with a two-way coupling). The core functionality is written in C, simulation is set up through a python interface, and visualization is handled by either OpenGL (real time) or Python (post-processing).
- Used the software and analytical methods to investigate geological processes related to rock deformation and geodynamics (video example).
- Wrote 5 successful research funding projects between 2015 and 2022 for a total of 22.2 M¥.
- Wrote or co-wrote 16 peer-reviewed articles between 2013 and 2021 with >400 cumulated citations.

University of Mainz – Germany

Post-doctoral researcher *Dec 2014 – Dec 2015*

- Developed a library to model 3D geometries based on a multi-layer 2D vector-graphics image (SVG). The library is used to design the initial configurations of numerical simulations
- Numerical simulations on super-computer, remote data analysis and visualization of >10TB data.
- Taught finite-element and machine learning classes (regression, PCA, clustering). BSc., MSc. level.

University of Lausanne – Switzerland

PhD candidate *Oct 2010 – Nov 2014*

- Teaching assistant for the finite element and numerical method classes at the University of Lausanne and ETH Zürich
- Teaching assistant for structural geology and field mapping classes/excursions and Introduction to Geology at University of Lausanne and EPFL.
- Improved the simulation software based on Lagrangian finite element, e.g. wrote a custom interpolation for the remeshing step that leverages the unstructured grid triangular finite elements. It performed 10x faster than the built-in Matlab used previously.

Education

University of Lausanne – Switzerland *Nov 2014*
PhD in Earth Sciences

University of Grenoble – France *Jun 2010*
MSc. in Earth Sciences

Additional

Languages: French, English, Japanese (conversational).
Accolade: Named "**outstanding reviewer 2021**" by Geophysical Journal International.
WaiDatathon 2021 - leader of the first prize winning team (1500$)
Full stack web-development bootcamp at Le Wagon - Tokyo (Jan. - Mar. 2022).
Blog on machine-learning including a reimplementation of the neural style transfer algorithm (PyTorch).
Academic publications (Google Scholar, PDFs).